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Thermodynamic analysis of Fe-Al-O system using Redlich-Kister (R-K) polynomial and Wagner's Taylor series expansion

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ABSTRACT

Solution behaviour can be described by determining the activity coefficients of its components. The decrease or increase of an activity depends on attractive or repulsive interactions between atoms/molecules in the solution. In the present work, activity coefficients (Al, O, Fe) and excess molar free energy of a ternary system (Fe-Al-O) have been calculated by employing quadratic and cubic formalisms using the compositional data obtained through Al-O equilibrium experiments in liquid iron. These activity coefficients have been compared with the experimentally (EMF) measured values reported in the literature by several researchers. The activity coefficients estimated using cubic formalism showed a good agreement with experimentally measured values. Drawing analogy with the steelmaking process, the activity coefficients were re-calculated for Fe-Al binary system, considering that the concentration of oxygen becomes insignificant in high Al steels, using two different approaches (a) R-K polynomial of the 3rd order and (b) Wagner's Taylor series expansion truncated at the 3rd order along with Gibbs-Duhem equation. The activity coefficients and excess molar free energy were found to be nearly same across all the aluminium concentration range using both these approaches. Finally, it can be concluded that Fe-Al-O ternary may well be treated as a Fe-Al binary system, beyond a deoxidation range (above 0.07 wt% Al), as the activity coefficients of Fe and Al were nearly the same, in both binary and ternary considerations.

1. Introduction

The activity coefficient of a solute in a ternary system can be represented by an infinite Taylor series as given by Wagner [1].

Symbolic representation of the nth differentials of $\ln \gamma_i$ with respect to (x_2) and (x_3) has been shown in Eq (1)

$$\begin{aligned}\ln \gamma_i(x_2, x_3) = \ln \gamma_i^o + (x_2\epsilon_i^{(2)} + x_3\epsilon_i^{(3)}) + (x_2)^2\rho_i^{(2)} + (x_3)^2\rho_i^{(3)} \\ + x_2x_3\,\rho_i^{(2,3)} + (x_2)^3\theta_i^{(2)} + \theta_i^{(3)}(x_3)^3 + (x_2)^2(x_3)\theta_i^{(2,2,3)} \\ + (x_2)(x_3)^2\theta_i^{(2,3,3)} + (x_2)^2(x_3)^2\alpha_i^{(2,2,3,3)} + (x_2)^3(x_3)\alpha_i^{(2,2,2,3)} \\ + (x_2)(x_3)^3\alpha_i^{(2,3,3,3)} + (x_2)^4\alpha_i^{(2)} + (x_3)^4\alpha_i^{(3)} + - - - \qquad (1)\end{aligned}$$

where γ_i denote the activity coefficients of solute 2 or solute 3, x_2 and x_3 are mole fractions of solute 2 and solute 3 respectively, γ_i^o is the activity coefficient obtained by the slope of the tangent drawn at infinite dilution on the curve plotted between activity and mole fraction of solute. $\epsilon_i^{(j)}$, $\rho_i^{(j,k)}$, $\theta_i^{(j,k,l)}$ and $\alpha_i^{(j,k,l,m)}$ represent the 1st, 2nd, 3rd and 4th order interaction parameters respectively.

The nature of interaction among the solute atoms in a solution can either increase or decrease their activities, for instance, in a typical Fe-Al-O (ternary) system, the strong attraction between solutes (Al and O), because of a large difference in their electronegativity values, decrease their individual activity values. The excess molar free energy of a ternary system (g^E) as a function of temperature, composition and the activity coefficient is given by Eq. (2) at a constant pressure. Moreover, activity coefficient is also a function of composition and the temperature.

$$g^E = RT(x_1\ln\gamma_1 + x_2\ln\gamma_2 + x_3\ln\gamma_3) \qquad (2)$$

where γ_1, γ_2 and γ_3 denote activity coefficients of solvent 1, solute 2 and solute 3 respectively on Raoultian scale. 'R' is the gas constant and 'T' is the temperature in Kelvin.

An analogy between the Taylor's series and the polynomial representing the activity coefficient (obtained by partial molar differentiation of excess molar free energy) yields a relationship between dimensionless coefficients and the interaction parameters [2]. So, the interaction parameters can be obtained by knowing the values of dimensionless coefficients [2,3]. The higher order interaction parameters can be directly

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calculated by estimating/knowing the lower order ones by considering the thermodynamic consistency. The dimensionless coefficients in the polynomial can either be obtained by multiple linear regression if excess molar free energy values (g^E) are known at each composition or through the equilibrium composition of solutes involved in a chemical reaction. For instance, by knowing the values of g^E at specific mole fraction points for any two components within the ternary system, one can obtain the values of all three, seven, and twelve dimensionless coefficients within quadratic, cubic, and quartic formulations respectively [2]. Likewise, knowing the equilibrium composition of solutes involved in a chemical reaction enables the determination of dimensionless coefficients [3]. The present calculations are based on WIPF and does not consider the pair formation like in quasichemical model. The activity coefficient in WIPF is a function of composition [4].

Galvanic cells can measure the activities of solutes in a solution through electromotive force. The Nernst equation relates the activity of a component (solute) at a specific temperature and concentration to the electromotive force (EMF). Consequently, the activity of a solute can be directly determined by calculating the measured voltage. A few researchers, specifically, Fruehan [5], Janke and Fischer [6] and Dimitrov [7] have reported directly measured activity values for the ternary system (Fe-Al-O). In the present work, the composition (Al and O) was obtained by the chemical analysis of the samples taken from the equilibrium melting experiments at 1873 K [3]. It should be noted here that the chemical analysis of the samples have been carried out for the total and dissolved oxygen contents [3] separately. Quantitative analysis of aluminium was done by using inductively coupled plasma atomic emission spectroscopy (ICP-AES) while dissolved and total oxygen were measured using two-stage heating method in inert-gas fusion absorptiometry. In two-stage heating method, sample was first heated to a temperature of 1873 K in a graphite crucible where all the dissolved oxygen is liberated. In the 2nd stage, the sample was heated to a temperature of 2613 K, where carbothermic reduction of alumina inclusions happened yielding the insoluble oxygen (oxygen present as alumina inclusions). This has been demonstrated in our previous work [3]. However, only the total oxygen which is the sum of dissolved oxygen and the insoluble oxygen, was considered in the present work. This is because the insoluble oxygen (oxygen present in the form of primary oxides) content was very small in the samples due to the unique sampling method employed in the study. This has been explained in detail in our previous work [3]. All the results in the present work are applicable for a liquid solution of Fe-Al-O or Fe-Al at 1873 K, though only the solid samples were investigated. The activity coefficients of solutes (Al and O) and solvent (Fe) have been expressed in terms of the dimensionless coefficients based on quadratic and cubic formalims. The need to go for higher order interaction parameter formalism is to make higher order polynomials applicable for assessment of concentrated solutions. In our previous manuscript, we have explicitly demonstrated that the experimental data at higher aluminium concentration was in good agreement with cubic formalism as compared to the quadratic one. The activity coefficients expressed in terms of dimensionless coefficients were substituted in the expression derived for the equilibrium constant of a chemical reaction. The values of some dimensionless coefficients have been obtained using multiple linear regression by using previously reported data from the literature [3,8,9] to improve the accuracy. The values of dimensionless coefficients can also be resubstituted in the original expression of $\ln\gamma_i$ to obtain numerical value of g^E at a given composition. Addressing the solution behaviour of aluminium and oxygen in liquid steel is crucial for steelmakers, as it significantly impacts the chemistry of steel. The activity coefficient of a solute reflects its stability within a solution and provides an estimate of the excess molar free energy. The present study compares two approaches for determining activity coefficients: direct EMF (electromotive force) measurements based on the Nernst equation, and calculations derived from equilibrium composition data. The key value of this work lies in estimating activity coefficients at high temperatures, such as those typical in steelmaking, without relying on direct EMF measurements. While measuring activity coefficients in organic compounds is relatively simple due to the lower temperatures involved, experimental measurement in steelmaking, at approximately 1873 K, is far more challenging. Therefore, this study focuses on estimating activity coefficients across a wide range of aluminium concentrations using equilibrium composition data.

In the present work, we have endeavored to estimate the change in activity coefficients of solute and solvent with solute concentration using two different formalisms. Additionally, the excess molar free energy change has also been plotted, considering ternary and binary system separately, comprising aluminium and oxygen as two solutes in solvent iron. The comparison of activity coefficient based on EMF measurement (experimental) and using compositional data has been elucidated for the first time. Additionally, the activity coefficients of solvent Fe have been calculated using two different approaches: (i) a R-K polynomial of 2nd and 3rd order and (ii) by applying Gibbs-Duhem equation to Taylor series expansion. However, the Gibbs-Duhem equation does not yield a polynomial and this method can be used only by considering a binary alloy consisting of a solvent and a solute. Interestingly, we have found matching results in the concentrated range by using both approaches, namely, R-K polynomial to a ternary system (Fe-Al-O system) and Gibbs-Duhem equation to a binary system (Fe-Al sytem). This has been explained in the Discussion section.

2. Example (Fe-Al-O system)

In this section, an example of a ternary system comprising Fe, Al and O is validated.

For a chemical reaction

$$2[Al] + 3[O] = Al_2O_{3\,(solid)} \tag{3}$$

Activity of alumina has been taken as unity (considering it pure)

The deoxidation constant K (inverse of equilibrium constant) can be written as

$$K = (a_{Al})^2 (a_O)^3 \tag{4}$$

$$2\ln\gamma_{Al} + 2\ln x_{Al} + 3\ln\gamma_O + 3\ln x_O = \ln K \tag{5}$$

where γ_{Al} and γ_O represent the activity coefficients of aluminum and oxygen while x_{Al} and x_O denote the mole fractions of aluminum and oxygen respectively.

Using Darken's quadratic formalism, Eqs. (6)–(8) can be obtained [2,3,10]

$$\ln\gamma_{Al} = \omega_0^{Fe,Al} - 2\omega_0^{Fe,Al}(x_{Al}) - \left(\omega_0^{Fe,Al} + \omega_0^{Fe,O} - \omega_0^{Al,O}\right)x_O + \omega_0^{Fe,Al}(x_{Al})^2 + \omega_0^{Fe,O}(x_O)^2 + (\omega_0^{Fe,Al} + \omega_0^{Fe,O} - \omega_0^{Al,O})x_{Al}x_O + C_1 \tag{6}$$

$$\ln\gamma_O = \omega_0^{Fe,O} - 2\omega_0^{Fe,O}(x_O) - \left(\omega_0^{Fe,Al} + \omega_0^{Fe,O} - \omega_0^{Al,O}\right)x_{Al} + \omega_0^{Fe,Al}(x_{Al})^2 + \omega_0^{Fe,O}(x_O)^2 + (\omega_0^{Fe,Al} + \omega_0^{Fe,O} - \omega_0^{Al,O})x_{Al}x_O + C_2 \tag{7}$$

$$\ln\gamma_{Fe} = \omega_0^{Fe,Al}(x_{Al})^2 + \omega_0^{Fe,O}(x_O)^2 + (\omega_0^{Fe,Al} + \omega_0^{Fe,O} - \omega_0^{Al,O})x_{Al}x_O \tag{8}$$

So, the activity coefficients γ_{Al} and γ_O both vary as a function of dimensionless coefficients ($\omega_k^{i,j}$) and the composition variables x_{Al} and x_O

At 1873 K, the values of $\omega_0^{Fe,Al}$, $\omega_0^{Fe,O}$, $\omega_0^{Fe,Al}+C_1$ and $\omega_0^{Fe,O}+C_2$ have been taken as −4.6095 [8], −9.548 [9], −3.8632 [5] and −4.568 [9] respectively.

On the basis of the equilibrium composition of aluminium and total oxygen in iron at 1873 K, the value of $\omega_0^{Al,O}$ has been estimated as −29.0123 [3] using multiple linear regression. As already mentioned, the term 'total oxygen' has been employed here in place of 'oxygen'

comprising the Fe-Al-O ternary system. The absolute values of dimensionless coefficients obtained using multiple linear regression for (i) Al vs total oxygen and (ii) Al vs dissolved oxygen were not found to be much different from each other [3].

Substituting values of all these dimensionless coefficients in Eq (6), Eq (7) and Eq (8), the plots of $\ln\gamma_i$ where 'i' represent either solute Al, O or solvent Fe, vs mole fraction of aluminium (on log plot) is obtained (see Fig. 1). In Fig. 1, the values of $\ln\gamma_{Fe}$, $\ln\gamma_{Al}$, $\ln\gamma_O$ are based on Al-O equilibria in Fe obtained from the study [3]. These activity coefficient values are based on the quadratic formalism. The values of $\ln\gamma_{Al}$ and $\ln\gamma_O$ have been taken from the work of Fruehan [5], Janke and Fischer [6] and Dimitrov [7]. The activity of oxygen was measured on a wt.% scale (using the EMF) so the $\ln\gamma_O$ values have been calculated using relationship as shown by Lupis [11]. This connects a relationship between Raoultian (mole fraction) to Henrian (mass%) standard state as shown by Eq (9). The activity coefficient values derived from directly measured activities of Al and O (Fruehan, Janke and Fischer and Dimitrov) are limited to only dilute region.

$$2.3\log f_i = \ln\gamma_i - \ln\gamma_i^o + \ln(x_{Fe}M_{Fe} + x_iM_i) - \ln(M_{Fe}) \tag{9}$$

where f_i denotes the activity coefficient of i on 1 wt% Henrian standard state and γ_i denotes the activity coefficient of i on Raoultian scale. γ_i^o is the activity coefficient of aluminium or oxygen at infinite dilution in solvent Fe and its value can be taken as exponential of $(\omega_i^{Fe,i} + C_2)$ where 'i' represent Al or O. M_{Fe} denote the atomic weight of iron and its value has been taken as 55.85.

Considering that the present study is focused in the high aluminium content range, it is important to invoke higher order formalisms, such as cubic formalism, for better representation of activity coefficients. The expressions for $\ln\gamma_{Fe}$, $\ln\gamma_{Al}$, and $\ln\gamma_O$ based on cubic formalism can be written as [3]

The plots of $\ln\gamma_{Fe}$, $\ln\gamma_{Al}$ and $\ln\gamma_O$ vs $\log x_{Al}$ have been obtained for cubic formalism as shown in Fig. 2, using the same methodology as employed for the quadratic formalism (Fig. 1). The values of dimensionless coefficients in Eq. (10–12) have been taken from Table 1.

A plot of oxygen activity (on a wt.% standard scale) versus wt.% aluminium (both axes on a logarithmic scale) shows a straight line with an intercept of −1.8 up to 1 wt% aluminium [4,12] Beyond this point, the curve exhibits an inward bend, indicating significant deviation in the concentrated range. By converting the wt.% standard state to a Raoultian standard state and expressing wt.% as mole fraction, we obtained a closer match with our calculated values especially in the high aluminium concentration range.

Excess molar free energy (g^E) is a function of composition and the natural logarithm of activity coefficient of the components (Eq. (2)) present in a ternary solution. The excess molar free energy is plotted in Fig. 3 considering 2nd order (quadratic) and 3rd order (cubic) polynomials (Eq (6)–(8) and Eq (10)–(12)). The excess molar free energy has also been calculated using the direct activity measurement data reported by Fruehan [5] and Dimitrov [7]. However, the value of $\ln\gamma_{Fe}$, required to calculate g^E, has not been reported in their work. The expression for $\ln\gamma_{Fe}$ is dependent on the dimensionless coefficients and variables (mole fractions of Al and O). So, as per the current study, values of $\ln\gamma_{Fe}$ have been calculated using Darken's quadratic formalism and then it has been substituted in Eq. (2) to obtain the values of g^E at a particular equilibrium composition of Al and O in iron.

3. Discussion

The natural logarihm of activity coefficients in a ternary system can be expressed by the nth order polynomial which can be expressed (i) using an indirect approach, that is, by partially differentiating the Redlich-Kister (R-K) polynomial which comprises the dimensionless coefficients and represents the excess molar free energy and (ii) using a direct approach, that is, by expressing the truncated Taylor's series in Wagner's interaction parameter formalism (WIPF) along with the special relationships obtained between the interaction parameters by applying Gibbs–Duhem equation. In the interaction parameter formalisms, the expression for $\ln\gamma_{solvent}$ can also be represented by a Taylor series comprising the similar interaction parameters as in $\ln\gamma_{solute}$. The higher order interaction parameters are rarely reported/available in the literature. However, using the analogy between the polynomials representing activity coefficient obtained by using (i) the indirect and (ii) the direct approach, as explained above, the higher order interaction parameters can be obtained from the dimensionless coefficients [2,3]. Further, such approach reduces the number of interaction parameters

$$\begin{aligned}\ln\gamma_{Fe} = {} & \left(\omega_0^{Fe,\ Al} + 3\omega_1^{Fe,Al}\right)(x_{Al})^2 + \left(\omega_0^{Fe,O} + 3\omega_1^{Fe,O}\right)(x_O)^2 + \left(\omega_0^{Fe,Al} + 2\omega_1^{Fe,Al} + \omega_0^{Fe,O} + 2\omega_1^{Fe,O} - \omega_0^{Al,O} - \omega^{Fe,Al,O}\right)x_{Al}x_O - (4\omega_1^{Fe,Al})(x_{Al})^3 - (4\omega_1^{Fe,O})(x_O)^3 \\ & + \left(-6\omega_1^{Fe,Al} - 2\omega_1^{Fe,O} - 2\omega_1^{Al,O} + 2\omega^{Fe,Al,O}\right)(x_{Al})^2x_O + \left(-2\omega_1^{Fe,Al} - 6\omega_1^{Fe,O} + 2\omega_1^{Al,O} + 2\omega^{Fe,Al,O}\right)x_{Al}(x_O)^2\end{aligned} \tag{10}$$

$$\begin{aligned}\ln\gamma_{Al} = {} & \omega_0^{Fe,\ Al} + \omega_1^{Fe,Al} + \left(-2\omega_0^{Fe,\ Al} - 6\omega_1^{Fe,Al}\right)x_{Al} \\ & + \left(-\omega_0^{Fe,\ Al} - 2\omega_1^{Fe,Al} - \omega_0^{Fe,O} - 2\omega_1^{Fe,O} + \omega_0^{Al,O} + \omega^{Fe,Al,O}\right)x_O \\ & + \left(\omega_0^{Fe,\ Al} + 9\omega_1^{Fe,Al}\right)(x_{Al})^2 \\ & + \left(\omega_0^{Fe,O} + \omega_1^{Fe,Al} + 6\omega_1^{Fe,O} - \omega_1^{Al,O} - \omega^{Fe,Al,O}\right)(x_O)^2 \\ & + \left(\omega_0^{Fe,\ Al} + 8\omega_1^{Fe,Al} + \omega_0^{Fe,O} + 4\omega_1^{Fe,O} - \omega_0^{Al,O} + 2\omega_1^{Al,O} - 3\omega^{Fe,Al,O}\right)x_{Al}x_O \\ & - (4\omega_1^{Fe,Al})(x_{Al})^3 - (4\omega_1^{Fe,O})(x_O)^3 \\ & + \left(-6\omega_1^{Fe,Al} - 2\omega_1^{Fe,O} - 2\omega_1^{Al,O} + 2\omega^{Fe,Al,O}\right)(x_{Al})^2x_O \\ & + \left(-2\omega_1^{Fe,Al} - 6\omega_1^{Fe,O} + 2\omega_1^{Al,O} + 2\omega^{Fe,Al,O}\right)x_{Al}(x_O)^2\end{aligned} \tag{11}$$

$$\begin{aligned}\ln\gamma_O = {} & \omega_0^{Fe,O} + \omega_1^{Fe,O} + \left(-\omega_0^{Fe,\ Al} - 2\omega_1^{Fe,Al} - \omega_0^{Fe,O} - 2\omega_1^{Fe,O} + \omega_0^{Al,O} + \omega^{Fe,Al,O}\right)x_{Al} + \left(-2\omega_0^{Fe,O} - 6\omega_1^{Fe,O}\right)x_O \\ & + \left(\omega_0^{Fe,\ Al} + 6\omega_1^{Fe,Al} + \omega_1^{Fe,O} + \omega_1^{Al,O} - \omega^{Fe,Al,O}\right)(x_{Al})^2 + \left(\omega_0^{Fe,O} + 9\omega_1^{Fe,O}\right)(x_O)^2 \\ & + \left(\omega_0^{Fe,Al} + 4\omega_1^{Fe,Al} + \omega_0^{Fe,O} + 8\omega_1^{Fe,O} - \omega_0^{Al,O} - 2\omega_1^{Al,O} - 3\omega^{Fe,Al,O}\right)x_{Al}x_O - (4\omega_1^{Fe,Al})(x_{Al})^3 - (4\omega_1^{Fe,O})(x_O)^3 \\ & + \left(-6\omega_1^{Fe,Al} - 2\omega_1^{Fe,O} - 2\omega_1^{Al,O} + 2\omega^{Fe,Al,O}\right)(x_{Al})^2x_O + \left(-2\omega_1^{Fe,Al} - 6\omega_1^{Fe,O} + 2\omega_1^{Al,O} + 2\omega^{Fe,Al,O}\right)x_{Al}(x_O)^2\end{aligned} \tag{12}$$

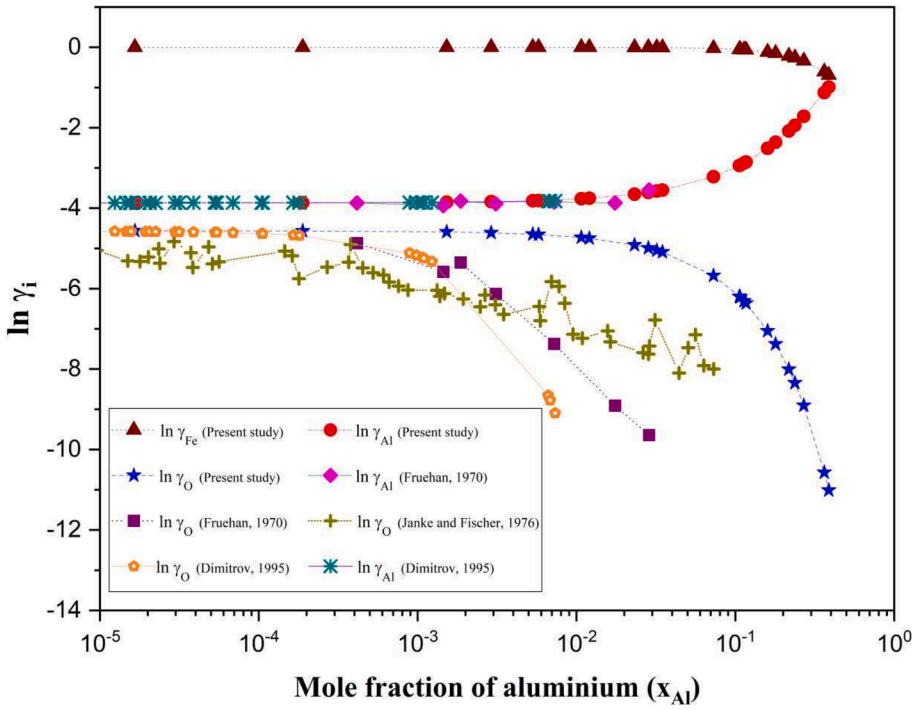


Fig. 1. Comparison of activity coefficients of solutes Al and O. The calculated activity coefficients in the present study is based on quadratic formalism using our equilibrium compositional data [3]. The comparison has been done with the experimental data (measured using EMF) of Fruehan [5], Janke and Fischer [6] and Dimitrov [7].

used to describe activity coefficients of solutes in a ternary system in a nth order truncation of Wagner's Taylor series from n^2+3n to $\frac{1}{2}(n-1)(n+4)$ [2].

In Fig. 1 and Fig. 2, $\ln\gamma_{Fe}$ do not change too much in the dilute range in both quadratic and cubic formalisms. It remains nearly flat except at very high aluminium concentration where it plunges down slowly. It can also be seen that $\ln\gamma_{Fe}$ approaches $\ln\gamma_{Al}$ closely on using quadratic formalism while there remains a gap in the cubic formalism. The value of $\ln\gamma_{Al}$ increases in the high aluminium region. However, this increase is gradual in both the formalisms. On the contrary, $\ln\gamma_O$ value plunges down steeply in both the formalisms. The minimum value of $\ln\gamma_O$ (-11)

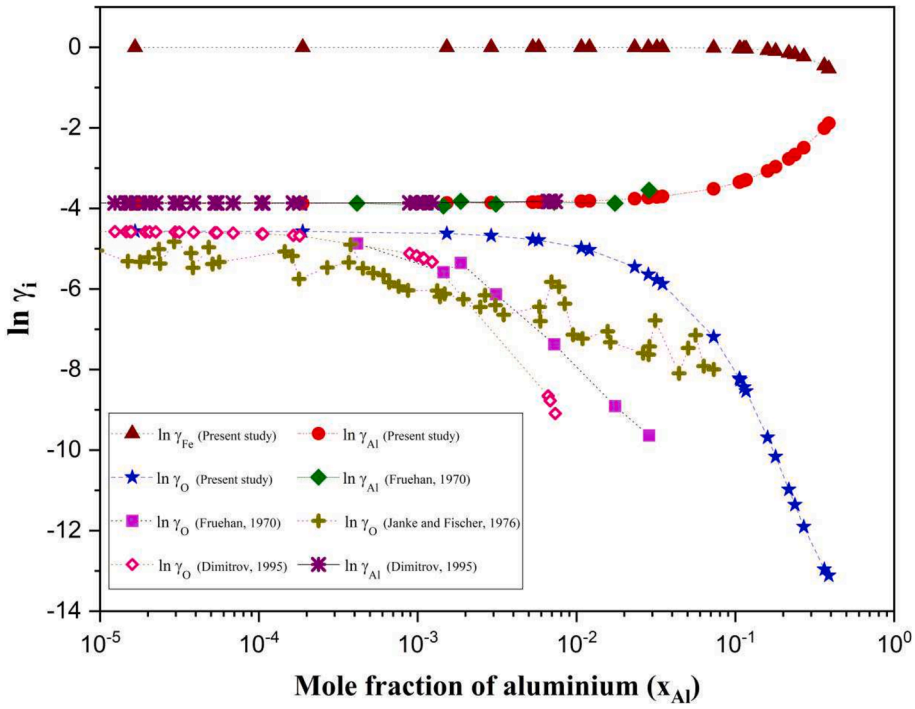


Fig. 2. Comparison of activity coefficients of solutes Al and O. The calculated activity coefficients in the present study is based on cubic formalism using our equilibrium compositional data [3]. The comparison has been done with the experimental data (measured using EMF) of Fruehan [5], Janke and Fischer [6] and Dimitrov [7].

Table 1
Dimensionless coefficients used in cubic formalism.

Parameter	Numerical value	Reference
$\omega_0^{Fe,\ Al}$	−4.6095	[8]
$\omega_1^{Fe,\ Al}$	0.741	[8]
$\omega_0^{Fe,\ O}$	−9.548	[9]
$\omega_1^{Fe,\ O}$	4.9872	[9]
$\omega_0^{Al,O}$	−29.0123	[3]
$\omega_1^{Al,O}$	37.1848	[3]
$\omega^{Fe,Al,O}$	−3.8613	[3]

in quadratic as shown in Fig. 1 differs considerably in cubic (−13.1) as shown in Fig. 2. Both the minimum values correspond to the highest aluminium concentration of x_{Al} (mole fraction of Al) as 0.387. In our previous work, we demonstrated that cubic formalism shows good agreement with experimental data in regions with high concentration of aluminium. The calculated $\ln\gamma_O$ values using cubic formalism, based on compositional data in the present work, align more closely with the experimental data measured using the EMF technique in Fig. 2 than in Fig. 1. This clearly indicates the precedence of cubic formalism over quadratic formalism.

Excess molar free energy (g^E) can be calculated using either the Redlich-Kister polynomials by substituting the values of dimensionless coefficients [2,3] by using Eq. (2). In the very dilute region, the g^E values are insignificant. This can be attributed to the lower mole fraction of solutes and also the negligible value of $\ln\gamma_{Fe}$ (which is close to zero). The $\ln\gamma_{Al}$ increases with increase in aluminium content which should increase the g^E. However, the term $x_{Al}\ln\gamma_{Al}$ (more negative) decreases as mole fraction of aluminium increases and also $x_{Fe}\ln\gamma_{Fe}$ term decreases which overall decrease the value of g^E.

The individual contribution by all the constituents comprising Fe-Al-O system in the excess molar free energy based on our compositional data is shown in Fig. 4.

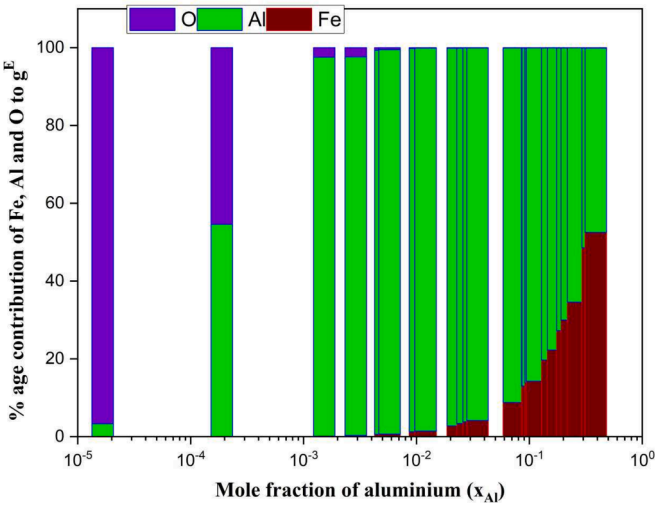


Fig. 4. Percentage contribution of each element (Fe, Al and O) to the excess molar free energy g^E (calculated using quadratic formalism) with increase in mole fraction of Al.

From Fig. 4, it is clear that the contribution of oxygen to the excess molar free energy diminishes quickly with aluminium addition. The value of excess molar free energy tends to be zero at infinite dilution and the contribution of iron is almost negligible. This is because the activity of iron tends to one when iron is almost pure. However, with aluminium addition, the magnitude of excess molar free energy increases and additionally, the presence of aluminium dominates the contribution to g^E for the entire concentration range. It is also worthy to mention that at higher aluminium concentration, that is, beyond mole fraction of 0.01, the contribution of iron on the g^E is noticeable. This can be well comprehended from Fig. 1 and Fig. 2 where the value of $\ln\gamma_{Fe}$ remains closer to one in the dilute region. It slowly descends as aluminium

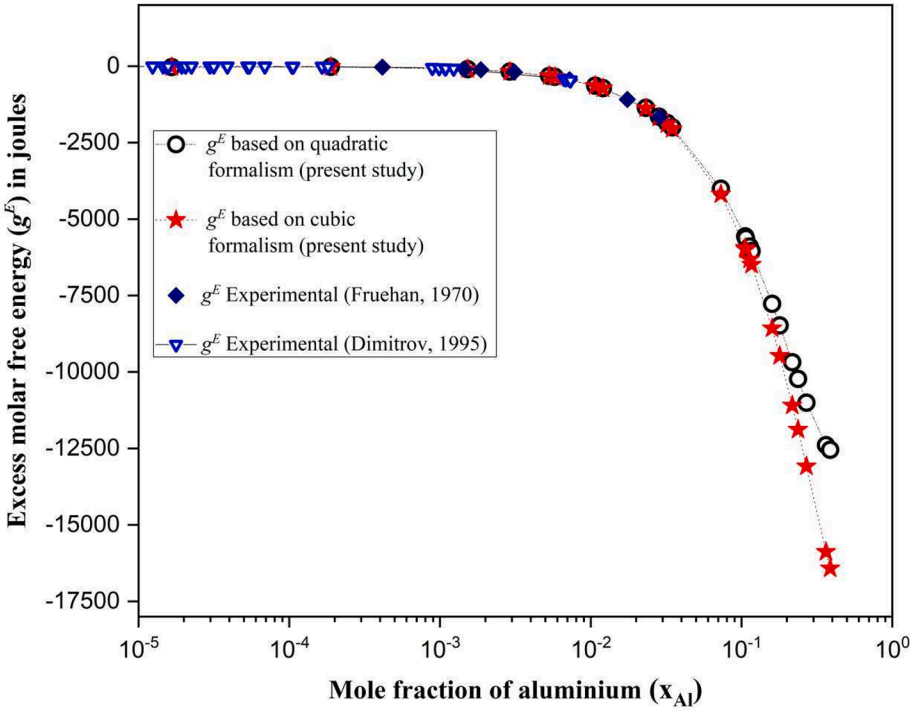


Fig. 3. Comparison of excess molar free energy (g^E) based on (i) quadratic and cubic formalisms using compositional data and (ii) experimentally measured values using EMF [5,7].

concentration increases. The presence of finite solute concentration shifts the activity of iron from one to the positive decimal number (less than 1) however, still insignificant as compared to the absolute value of $\ln\gamma_{Al}$ and $\ln\gamma_{O}$. The mole fraction of Fe is more than 0.6 in both the formalisms (Fig. 1 and Fig. 2). On the contrary the value of $\ln\gamma_{O}$ is far less than both $\ln\gamma_{Al}$ and $\ln\gamma_{Fe}$ along the whole concentration range of solute (Al).

It should be mentioned here that $\ln\gamma_{solvent}$ or $\ln\gamma_{Fe}$ in the expression for g^{E}(experimental) is based on the calculations using Eq. (8). This may seem questionable as we are taking only the experimental values of $\ln\gamma_{i}$. However, as already explained that (i) $\ln\gamma_{Fe}$ was not available and (ii) its individual value was not as significant as that of $\ln\gamma_{Al}$ and $\ln\gamma_{O}$.

3.1. Considering binary system (Fe-Al alloy)

In Fig. 4, it can be seen clearly that the contribution to the excess molar free energy is mainly from aluminium and iron. Nevertheless, at almost high dilution of Al, the contribution from oxygen is significant and it decays exponentially and becomes almost insignificant once aluminium concentration is perceptible. So, it will be fair assumption to consider iron and aluminium (with exclusion of oxygen) as the binary system for most of the experimental data. The activity coefficient of solute 2 (aluminium) and solvent (iron) can be obtained using two ways as given below.

3.1.1. Using R-K polynomial

Considering a quadratic polynomial, the excess free energy of a binary system (comprising Fe as solvent and Al as solute) (g^{E}) can be written as

$$\frac{g^{E}}{RT} = \omega_0^{12}\left[x_{Al} - (x_{Al})^2\right] \tag{13}$$

The expressions for $\ln\gamma_{solvent}$ and $\ln\gamma_{solute}$ can be obtained by multiplying Eq. (13) by total number of moles and then differentiating partially with respect to number of moles of Fe and Al.

$$\ln\gamma_{Fe} = \omega_0^{12}(x_{Al})^2 \tag{14}$$

$$\ln\gamma_{Al} = \omega_0^{12}(1 - x_{Al})^2 \tag{15}$$

Similarly, considering a cubic polynomial, the expression for (g^{E}) can be written as

$$\frac{g^{E}}{RT} = \left[x_{Al} - (x_{Al})^2\right]\left[\omega_0^{12} + \omega_1^{12}(1 - 2x_{Al})\right] \tag{16}$$

$$\ln\gamma_{Fe} = \omega_0^{12}(x_{Al})^2 + \omega_1^{12}\left[3(x_{Al})^2 - 4(x_{Al})^3\right] \tag{17}$$

$$\ln\gamma_{Al} = \omega_0^{12}(1 - x_{Al})^2 + \omega_1^{12}\left[-4(x_{Al})^3 + 9(x_{Al})^2 - 6x_{Al} + 1\right] \tag{18}$$

3.1.2. Using Taylor series expansion and Gibbs-Duhem equation

Considering the Taylor series expansion (1), the expression for $\ln\gamma_2$ can be written as

$$\ln\gamma_2(x_2) = \ln\gamma_2^{o} + (x_2)\epsilon_2^{(2)} + (x_2)^2\rho_2^{(2)} + (x_2)^3\theta_2^{(2)} + - - \tag{19}$$

The activity coefficient of iron can be obtained using the Gibbs-Duhem equation as:

$$(1 - x_2)d\ln\gamma_{solvent} + x_2 d\ln\gamma_2 = 0 \tag{20}$$

Using a 3rd order truncation of $\ln\gamma_2$ in Eq. (19) and putting its value in Eq. (20),

$$(1 - x_2)d\ln\gamma_{solvent} + x_2 d\left(\ln\gamma_2^{o} + x_2\epsilon_2^{(2)} + (x_2)^2\rho_2^{(2)} + (x_2)^3\theta_2^{(2)}\right) = 0 \tag{21}$$

$$\ln\gamma_{solvent} = -\epsilon_2^{(2)}\int\frac{x_2 dx_2}{(1 - x_2)} - 2\rho_2^{(2)}\int\frac{(x_2)^2 dx_2}{(1 - x_2)} - 3\theta_2^{(2)}\int\frac{(x_2)^3 dx_2}{(1 - x_2)} \tag{22}$$

$$\begin{aligned}\ln\gamma_{solvent} = {} & \epsilon_2^{(2)}\left[\ln(1 - x_2) + x_2 - 1\right] \\ & + 2\rho_2^{(2)}\left[\ln(1 - x_2) + \frac{(1 - x_2)^2}{2} + 2x_2 - 2\right] \\ & + 3\theta_2^{(2)}\left[\ln(1 - x_2) - \frac{(1 - x_2)^3}{3} + 3\frac{(1 - x_2)^2}{2} + 3x_2 - 3\right] + c\end{aligned} \tag{23}$$

At $x_2 = 0$, $\gamma_{solvent} = 1 and \ln\gamma_{solvent} = 0$

The integration constant for Eq. (2) is $c = \epsilon_2^{(2)} + 3\rho_2^{(2)} + \frac{11}{2}\theta_2^{(2)}$

$$\begin{aligned}\ln\gamma_{solvent} = {} & \epsilon_2^{(2)}\left[\ln(1 - x_2) + x_2\right] + 2\rho_2^{(2)}\left[\ln(1 - x_2) + \frac{(x_2)^2}{2} + x_2\right] \\ & + 3\theta_2^{(2)}\left[\ln(1 - x_2) + \frac{(x_2)^3}{3} + \frac{(x_2)^2}{2} + x_2\right]\end{aligned} \tag{24}$$

For a binary system, the activity coefficient of a solvent can be directly derived through integration, as mentioned above, by using the known activity coefficient expression of the solute. However, in a ternary system, there are two variables (the mole fractions of two different solutes). In such cases, a different approach has been used, involving path integration [13]. The activity coefficient of the solvent in a ternary system can also be determined by expressing it as an nth-order polynomial (Taylor series) and solving it using the Gibbs-Duhem equation [2].

Considering binary system (Fe-Al) and the 3rd order truncated Taylor series representing activity coefficients, $\ln\gamma_{Fe}$ and $\ln\gamma_{Al}$ can be expressed as

$$\begin{aligned}\ln\gamma_{Fe} = {} & \epsilon_{Al}^{(Al)}\left[\ln(1 - x_{Al}) + x_{Al}\right] + 2\rho_{Al}^{(Al)}\left[\ln(1 - x_{Al}) + \frac{(x_{Al})^2}{2} + x_{Al}\right] \\ & + 3\theta_{Al}^{(Al)}\left[\ln(1 - x_{Al}) + \frac{(x_{Al})^3}{3} + \frac{(x_{Al})^2}{2} + x_{Al}\right]\end{aligned} \tag{25}$$

$$\ln\gamma_{Al} = \ln\gamma_{Al}^{o} + (x_{Al})\epsilon_{Al}^{(Al)} + (x_{Al})^2\rho_{Al}^{(Al)} + (x_{Al})^3\theta_{Al}^{(Al)} \tag{26}$$

The values of interaction parameters $\epsilon_{Al}^{(Al)}$, $\rho_{Al}^{(Al)}$ and $\theta_{Al}^{(Al)}$ have been taken as 4.773, 2.06 and −2.964 respectively [3]. The value of $\ln\gamma_{Al}^{o}$ has been taken as −3.8685 [5]

In our previous work [3], we have shown that cubic formalism gives a good agreement with the experimental data particularly at high aluminium concentration. So, the values of activity coefficients $\ln\gamma_{Fe}$ and $\ln\gamma_{Al}$ have been compiled in Table 2 using different methods: cubic formalism considering ternary system (Eq. (10) & Eq. (11)), R-K polynomial considering binary system (Eq. (17) and Eq. (18)) and Taylor series expansion up to 3rd order with Gibbs-Duhem Equation (Eq. (25) and Eq. (26)).

The expression for $\ln\gamma_{solvent}$ as represented by Eq. (24) is now a different function. It is no more an ordinary n^{th} order polynomial. For obtaining this expression, the thermodynamic consistency has been preserved and the equation has been derived using Gibbs-Duhem equation considering binary Fe-Al system. It is interesting to note that the values of $\ln\gamma_{solvent}$ ($\ln\gamma_{Fe}$) obtained by considering ternary system differs considerably as compared to $\ln\gamma_{solvent}$ values obtained by considering a binary Fe and Al system in low aluminium concentration range. Though, the difference is not discernible in Fig. 5, it becomes evident from Table 2. The difference in $\ln\gamma_{Fe}$ values diminishes in the high aluminium concentration range and no noticeable difference is seen in the values of $\ln\gamma_{Fe}$ and $\ln\gamma_{Al}$ obtained using R-K polynomial of 3rd order considering binary alloy of Fe and Al (Eq. (17) and Eq. (18)) and using Taylor series truncated at 3rd order with Gibbs-Duhem

Table 2
Mole fraction of Al and O (TO) [3] and values of $\ln\gamma_{solvent}$ and $\ln\gamma_{Al}$ calculated by considering ternary system (Fe, Al and O) and binary system (Fe and Al) using different methods.

Mole fraction of Al(x_{Al})	Mole fraction of O(x_O)	$\ln\gamma_{Fe}$ for (Fe, Al, O) system using Eq. (10)	$\ln\gamma_{Al}$ for (Fe, Al, O) system using Eq. (11)	$\ln\gamma_{Fe}$ for (Fe, Al) system using Eq. (17)	$\ln\gamma_{Al}$ for (Fe, Al) system using Eq. (18)	$\ln\gamma_{Fe}$ for (Fe, Al) system using Eq. (25)	$\ln\gamma_{Al}$ for (Fe, Al) system using Eq. (26)
1.7E-05	4.16E-04	1.12E-06	−3.8848	−6.6E-10	−3.8684	−6.90E-10	−3.8684
0.0002	1.33E-04	6.68E-07	−3.8728	−8.5E-08	−3.8676	−8.62E-08	−3.8676
0.0015	3.18E-05	−4.27E-06	−3.8624	−5.5E-06	−3.8612	−5.52E-06	−3.8612
0.0029	5.79E-05	−1.57E-05	−3.8569	−2E-05	−3.8546	−2.01E-05	−3.8546
0.0053	2.97E-05	−6.32E-05	−3.8443	−6.7E-05	−3.8432	−6.72E-05	−3.8432
0.0058	2.25E-05	−7.66E-05	−3.8418	−8E-05	−3.8409	−8.00E-05	−3.8409
0.0107	1.87E-05	−2.72E-04	−3.8179	−0.0003	−3.8172	−2.77E-04	−3.8172
0.0120	1.16E-05	−3.47E-04	−3.8112	−0.0004	−3.8108	−3.51E-04	−3.8108
0.0232	2.42E-05	−0.0013	−3.7576	−0.0013	−3.7567	−0.0013	−3.7567
0.0283	2.7E-05	−0.0020	−3.7327	−0.0020	−3.7317	−0.0020	−3.7317
0.0320	1.52E-05	−0.0025	−3.7142	−0.0025	−3.7136	−0.0025	−3.7136
0.0347	1.75E-05	−0.0030	−3.7011	−0.0030	−3.7005	−0.0030	−3.7005
0.0729	3.21E-05	−0.0138	−3.5120	−0.0138	−3.5110	−0.0138	−3.5110
0.1058	5.41E-05	−0.0301	−3.3453	−0.0303	−3.3438	−0.0303	−3.3438
0.1070	5.2E-05	−0.0308	−3.3392	−0.0310	−3.3378	−0.0310	−3.3378
0.1133	4.1E-05	−0.0348	−3.3065	−0.0349	−3.3054	−0.0350	−3.3054
0.1166	3.91E-05	−0.0371	−3.2895	−0.0372	−3.2885	−0.0372	−3.2885
0.1594	5.22E-05	−0.0724	−3.0687	−0.0726	−3.0676	−0.0726	−3.0675
0.1791	4.72E-05	−0.0934	−2.9656	−0.0936	−2.9647	−0.0936	−2.9646
0.2169	7.12E-05	−0.1421	−2.7679	−0.1425	−2.7667	−0.1425	−2.7667
0.2368	5.53E-05	−0.1728	−2.6629	−0.1732	−2.6622	−0.1732	−2.6621
0.2694	3.12E-05	−0.2310	−2.4914	−0.2312	−2.4910	−0.2312	−2.4910
0.3631	7.87E-05	−0.4557	−2.0064	−0.4564	−2.0059	−0.4565	−2.0059
0.3872	7.62E-05	−0.5300	−1.8842	−0.5298	−1.8840	−0.5298	−1.8838

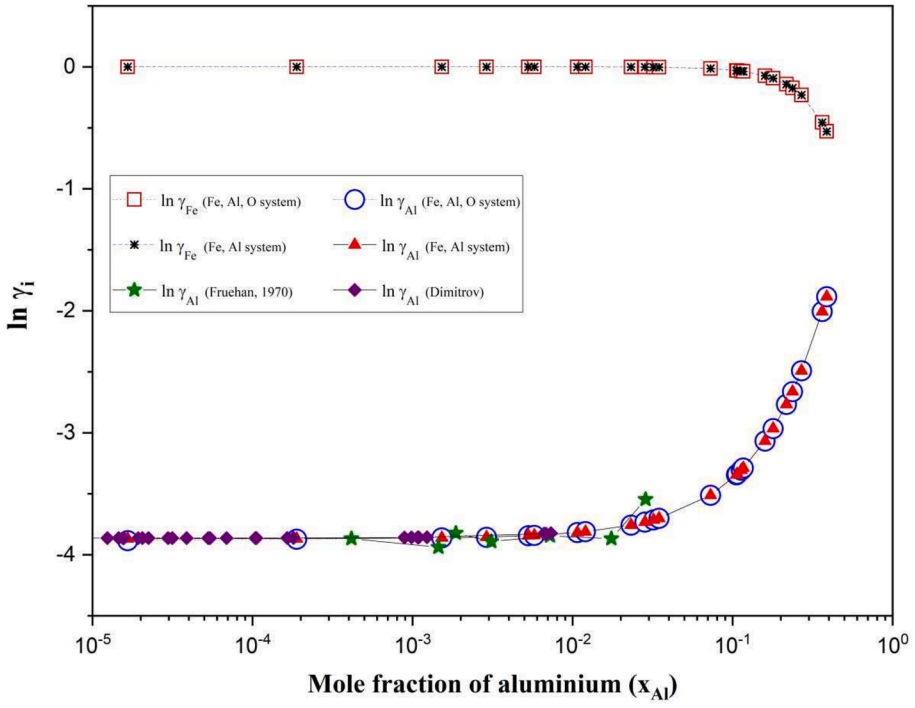


Fig. 5. Comparison of activity coefficients of solvent Fe and solute Al (calculated and experimentally measured [5,7]. The calculated activity coefficients in the present study are based on cubic formalism (considering ternary alloy) and 3rd order truncated Taylor series with Gibbs-Duhem equation considering binary alloy of Fe and Al.

equation (Eq. (25) and Eq. (26)).

The plots in Fig. 5 and Fig. 6 illustrate a comparison of activity coefficients and g^E, respectively, as functions of $\log x_{Al}$. This comparison is based on analysing the ternary alloy using cubic formalism and the binary alloy using a Taylor series truncated at the third order, incorporating the Gibbs-Duhem equation.

The trend for values of excess molar free energy (g^E) is similar to that of $\ln\gamma_{Fe}$ values. Considering a ternary system, both quadratic and cubic formalism yields the value of g^E as −30.59 J/mole however, if a binary system of Fe and Al is considered, the value is only −1 J/mole when mole fraction of aluminium is 1.7E-05. This can be comprehended from the fact that major contribution to g^E at very low aluminium concen-

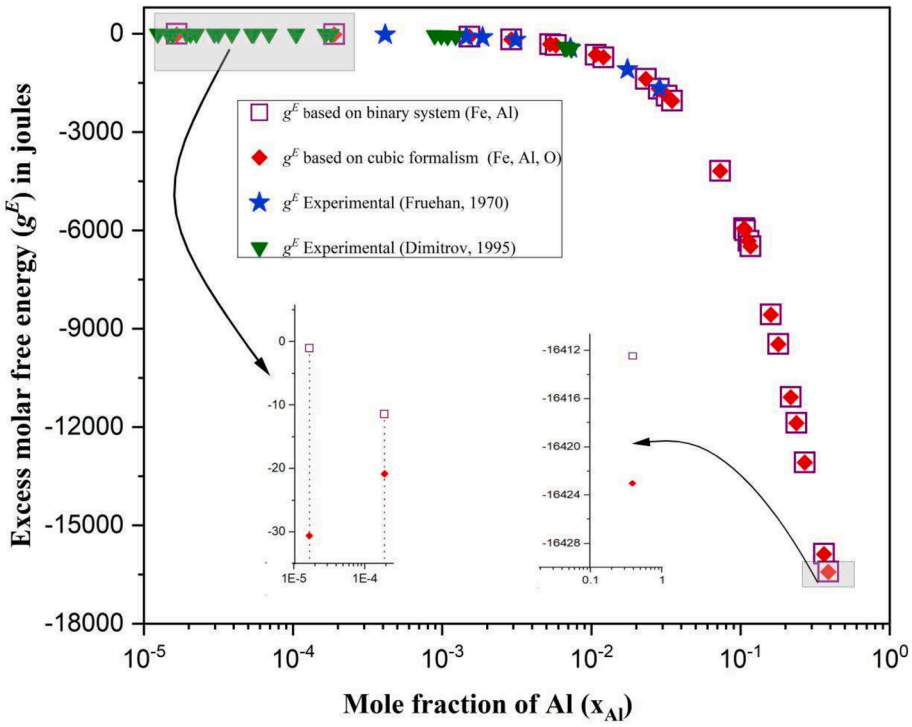


Fig. 6. Comparison of excess molar free energy (g^E) based on ternary (Fe, Al, O) using cubic formalism and binary alloy (Fe, Al) system using Taylor series of 3rd order and Gibbs-Duhem equation. The values of g^E has been plotted against mole fraction of Al (on log scale).

tration is from oxygen as can be seen from Fig. 4. The distinction remains imperceptible in the plot depicted in Fig. 6, primarily due to significant margins in the divisions of the y-axis. When mole fraction of aluminium is 0.387, the value of g^E is −16423 J/mole considering ternary system and cubic formalism while it is −16412.47 J/mole when a binary system is considered.

In the present study, $\ln\gamma_i$ and g^E have been calculated for the aluminium concentration (in mole fraction) ranging from 0.0000166 to 0.387.

3.1.3. *Implications on excess molar free energy and excess stability*

Since the excess molar free energy of the binary system (Fe-Al) can be written as (by substituting Eq. (25) and Eq. (26) in Eq. (2))

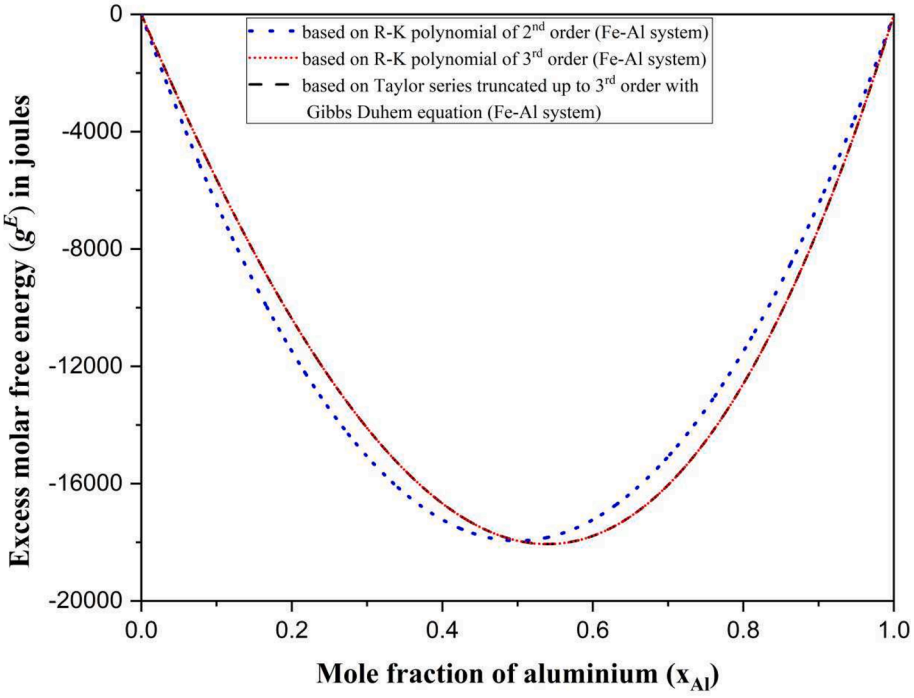


Fig. 7. Plot of excess molar free energy g^E as a function of mole fraction of aluminium using different methods considering binary system (Fe-Al).

$$\frac{g^E}{RT} = x_{Al}\Big[\ln\gamma^o_{Al} + (x_{Al})\epsilon^{(Al)}_{Al} + (x_{Al})^2\rho^{(Al)}_{Al} + (x_{Al})^3\theta^{(Al)}_{Al}\Big] + (1-x_{Al})\Bigg[\epsilon^{(Al)}_{Al}\{\ln(1-x_{Al}) + x_{Al}\} + 2\rho^{(Al)}_{Al}\left\{\ln(1-x_{Al}) + \frac{(x_{Al})^2}{2} + x_{Al}\right\} + 3\theta^{(Al)}_{Al}\left\{\ln(1-x_{Al}) + \frac{(x_{Al})^3}{3} + \frac{(x_{Al})^2}{2} + x_{Al}\right\}\Bigg] \tag{27}$$

The values of $\epsilon^{(Al)}_{Al}, \rho^{(Al)}_{Al}, \theta^{(Al)}_{Al}$ have been taken as $4.773, 2.06$ and -2.964 respectively [3].

Eq. (13) is maximized/minimized at $x_{Al} = 0.5$ regardless of the value of ω_0^{12} as it is evident that quadratic one is a symmetric function while the critical points of Eq. (16) are obtained by differentiating it with respect to x_{Al} and the values of x_{Al} are given by the expression given in Eq. (28)

$$x_{Al} = \frac{(2\omega_0^{12} + 6\omega_1^{12}) \pm \sqrt{(2\omega_0^{12} + 6\omega_1^{12})^2 - 24\omega_1^{12}(\omega_1^{12}+\omega_1^{12})}}{12\omega_1^{12}} \tag{28}$$

The value of x_{Al} in Eq. (28) is dependent on dimensionless coefficients. Similarly, for Eq. (27), the maxima/minima of the function depend on the dimensionless coefficients.

Minimizing the function g^E based on Eq. (27) (by differentiating it with respect to the mole fraction of aluminum), we find that the solution can be obtained numerically. The function reaches its minimum at $x_{Al} = 0.5394$, with g^E attaining a value of -18058.82J/mole at this composition (by substituting the values of interaction parameters and temperature of 1873 K). It is worth noting that unlike the configurational entropy or enthalpy of mixing in a regular solution, the plot for cubic formalism does not exhibit symmetry around a mole fraction of 0.5. At both ends, it is quite evident that g^E is zero when any of the pure component is taken. However, in quadratic formalism, the function is minimized at $x_{Al} = 0.5$. At $x_{Al} = 0.5$, quadratic formalism is symmetric while no such symmetry is observed in cubic formalism.

A similar expression can be obtained for binary system (Fe-O)

$$\frac{g^E}{RT} = x_O\Big[\ln\gamma^o_O + (x_O)\epsilon^{(O)}_O + (x_O)^2\rho^{(O)}_O + (x_O)^3\theta^{(O)}_O\Big] + (1-x_O)\Bigg[\epsilon^{(O)}_O\{\ln(1-x_O) + x_O\} + 2\rho^{(O)}_O\left\{\ln(1-x_O) + \frac{(x_O)^2}{2} + x_O\right\} + 3\theta^{(O)}_O\left\{\ln(1-x_O) + \frac{(x_O)^3}{3} + \frac{(x_O)^2}{2} + x_O\right\}\Bigg] \tag{29}$$

The values of $\epsilon^{(O)}_O, \rho^{(O)}_O, \theta^{(O)}_O$ have been taken as $-10.827, 35.337$ and -19.949 respectively [3].

The minimum value in Fig. 8 has been observed at $x_O = 0.61$ and the corresponding value of g^E is -39436 J/mole. This is more than twice the value obtained in Fe-Al system (at $x_O = 0.5394$).

From Fig. 7, it is interesting to note that the plotted curves based on R-K polynomial of 3rd order shows difference, albeit marginally with R-K quadratic polynomial one but it does not show distinguishable difference with g^E values based on Eq (27) if binary system of Fe-Al is considered (the two curves completely merge with each other). The same is also evident from Fig. 8. It is to be noted that the plots in Fig. 7 and Fig. 8 look different than in Fig. 3 and Fig. 6 and it may seem contradicting. However, the scales are different. In Fig. 3 and Fig. 6, the x-axis is on log scale while in Fig. 7 and Fig. 8, it is a linear scale. Further, the scale covers the entire concentration range of aluminium in Fig. 7 and Fig. 8.

Excess stability in a binary system is determined by taking the double

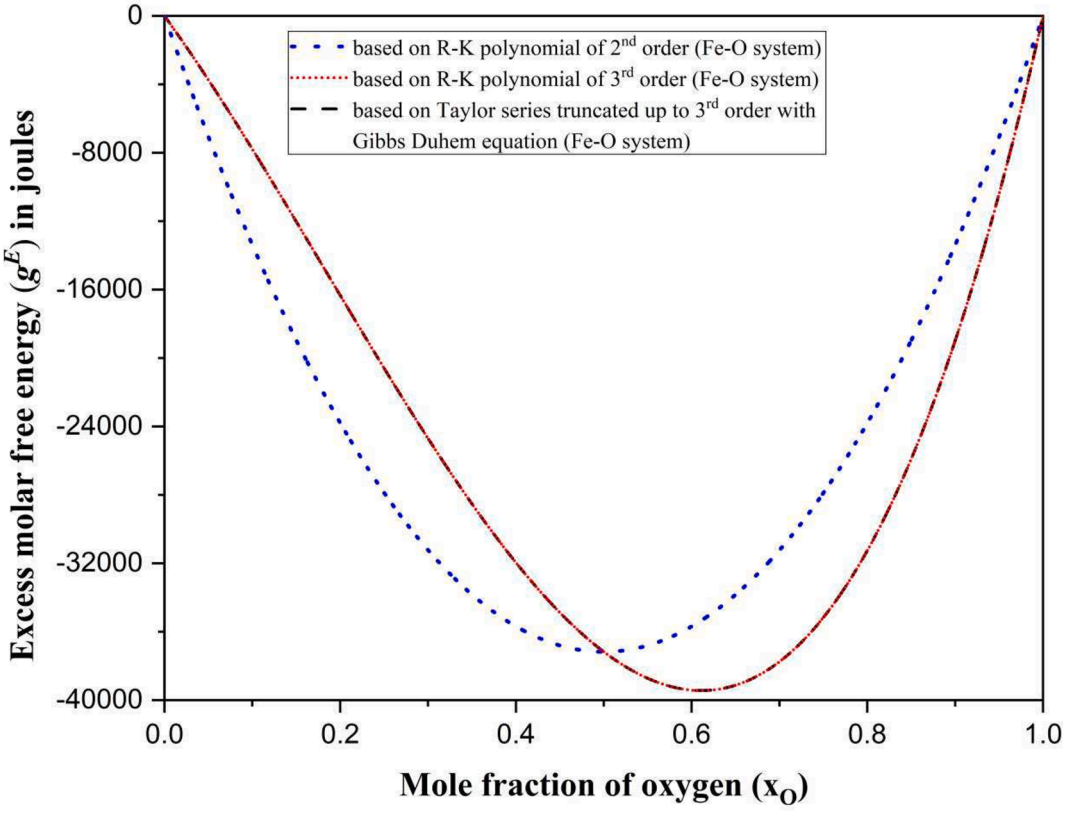
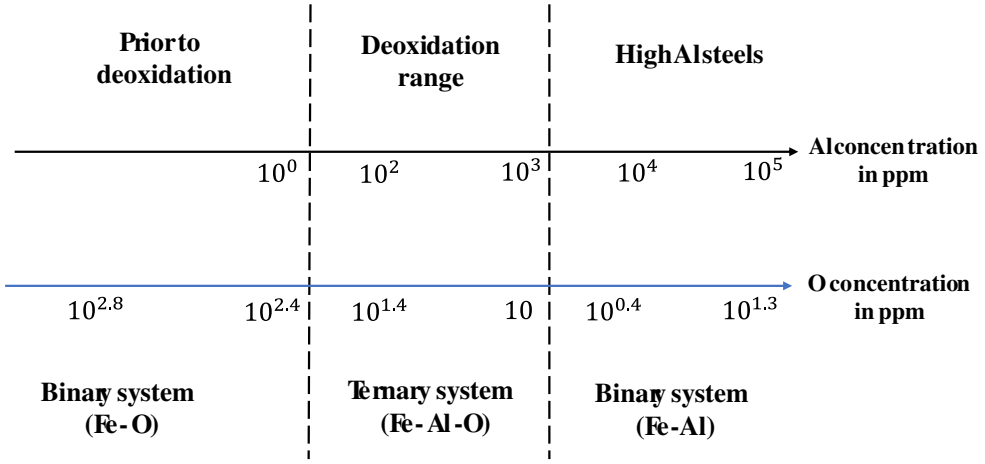


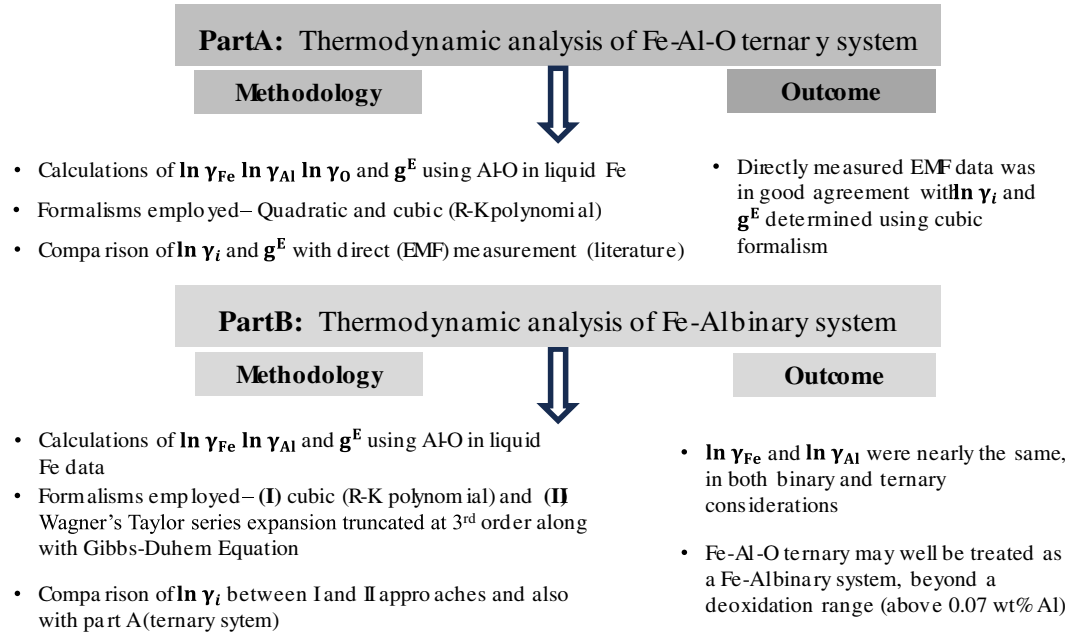
Fig. 8. Plot of excess molar free energy g^E as a function of mole fraction of oxygen using different methods considering binary system (Fe-O).

derivative of excess molar free energy $\left(\frac{d^2g^E}{dx_2^2}\right)$ or $f''g^E$ with respect to the mole fraction of solute [14]. For an ideal solution, the enthalpy of mixing is 0 while for regular solutions, it is non-zero. In both the cases, there is random arrangement of atoms/molecules in the solution. However, considering quasichemical model, there are formation of pairs due to a strong mutual interaction between the components of the pairs. So, in a quasichemical model, the solution cannot be termed as homogenous unlike ideal and regular solution. Similar phenomenon happens when there is formation of intermetallics/compounds and results in formation of two solutions differing only infinitesimally in composition [14]. The free energy sharply depicts a deep dip when intermetallics or stable compounds are formed at a typical stoichiometric composition. Consequently, the typical peaks are observed at a specific composition in the excess stability plot. Similarly, the binary plot of Fe-Al should exhibit peaks at compositions where intermetallic compounds form like Fe_3Al, FeAl, $FeAl_2$, Fe_2Al_5, $FeAl_3$, making these points discernible at room temperature. Using a quadratic polynomial for the excess molar free energy, it is clear that $\left(\frac{d^2g^E}{dx_2^2}\right)$ will be constant and will show a flat line parallel to the x-axis with an intercept of $RT(\omega_0^{Fe,\ Al})$ if binary system of Fe-Al is considered while the intercept would be $RT(\omega_0^{Fe,\ O})$ if binary system of Fe-O would be considered. Considering R-K polynomial of order 3, $f''g^E$ will show a straight line with slope of $12RT\omega_1^{Fe,\ Al}$ with an intercept of $-RT\left(2\omega_O^{Fe,\ Al}+6\omega_1^{Fe,\ Al}\right)$ for Fe-Al system while for Fe-O system, slope will be $12RT\omega_1^{Fe,\ O}$ with an intercept of $-RT\left(2\omega_O^{Fe,\ O}+6\omega_1^{Fe,\ O}\right)$.

For Eq. (27) and Eq. (29), the excess stability is given by



(a)



(b)

Fig. 9. (a) Categorization (schematic) of the Fe-Al-O system in relevance to various stages of steelmaking (b) Methodology adapted for thermodynamic analysis of a Fe-Al-O system under binary and ternary considerations.

$$\left(\frac{d^2g^E}{dx_2^2}\right) = \mathrm{RT}\left[\frac{\epsilon_j^j + 2\rho_j^j(x_2) + 3\theta_j^j(x_2)^2}{1 - x_2}\right] \quad (30)$$

where ϵ_j^j, ρ_j^j and θ_j^j represent the self − first, second and third order interaction parameters respectively. If we account for second-order truncation, the numerator in Eq. (30) will include only the first two terms, while the denominator will remain unchanged. Eq. (30) monotonically increases and is discontinuous at $x_2 = 1$. However, if we only consider the numerator portion, then function increases and then decreases showing a parabolic path while it will exhibit a straight line with slope of $2RT\rho_j^j$ considering 1st two terms. All the aforementioned equations are based on WIPF [2,3]. So, any peaks observed at a specific composition cannot be reproduced across a broad range of composition at ambient temperatures [15].

3.2. Relevance in Fe-Al-O system

In the present study, we have shown that oxygen tends to stay in liquid iron as long as no deoxidizer is added. Aluminium, as a solute, is also stable in liquid iron. However, with simultaneous presence of these two solutes in liquid iron, the stability varies substantially considering two separate (namely, Fe-O and Fe-Al) binary systems. The values of $\ln\gamma_{Al}^{o}$ and $\ln\gamma_{O}^{o}$ are both negative but 1st order self-interaction parameter ϵ_{Al}^{Al} is positive while ϵ_{O}^{O} is negative. This indicates that aluminium does not want its immediate neighbourhood with aluminium atoms while oxygen has considerable affinity towards oxygen atoms. This is a key factor contributing to the increase in the activity coefficient of aluminium when more aluminium is added to liquid iron. The curves plotted for g^E values for Fe-Al and Fe-O system also show a variation. The g^E shows an inflection point when $\epsilon_j^j < O$ [15]. In the present study, there is a significant alteration in the concavity of the curve shown in Fig. 7 and Fig. 8.

Prior to the deoxidation process, the oxygen concentration typically falls within the range of 400–800 ppm, diminishing to just a few ppm in the case of deoxidized steel. Consequently, following the primary steelmaking, the molten steel can be regarded as a binary alloy (Fe-O system), assuming minimal presence of other impurities in the melt. At lower concentrations of aluminium, steel can be considered as a ternary system, with both aluminium and oxygen present at parts per million (ppm) levels, while iron (Fe) forms the major component in the system. In an alloying range of aluminium, the steel can be regarded as a binary system of Fe-Al as both components are in wt.% while oxygen content is insignificant. This categorization of the Fe-Al-O system is shown schematically in Fig. 9(a). The methodology adapted for the thermodynamic analysis of Fe-Al-O system under binary and ternary considerations is summarized shown in Fig. 9(b).

4. Conclusions

In this study, we have compared the activity coefficients of the solvent (Fe) and solutes (Al and O), determined on the basis of equilibrium compositional data for the Fe-Al-O system, using quadratic and cubic formalisms with directly measured (EMF experimental data) values available in the literature. Additionally, we compared the excess molar free energy values derived from these different formalisms. Thermodynamic analysis by employing suitable formalisms was also conducted, focusing on the binary Fe-Al system in a high aluminium range. Based on the present study, the following conclusions can be made:

1. As the concentration of aluminium increases, the activity coefficients of iron and oxygen decrease, whereas the activity coefficient of aluminium increases. However, the excess molar free energy (g^E) decreases steadily with aluminium concentration. The cubic formalism estimates the activity coefficients values much closer to that of experimental (EMF measurement) values.
2. The experimentally measured values of activity coefficients using EMF (from literature), especially $\ln\gamma_O$ shows good agreement with the values determined using cubic formalism (based on the compositional equilibrium data of Al-O in liquid iron).
3. The comparative analysis showed an excellent agreement between the activity coefficients ($\ln\gamma_i$) and excess molar free energy (g^E) estimated using cubic formalism under binary and ternary considerations for Al concentration above 0.07 wt%, that is, beyond a deoxidation range.
4. For a Fe-Al binary system, using (a) R-K polynomial of the third order (b) or Taylor series truncated at the third order along with the Gibbs-Duhem equation, the activity coefficients and the excess molar free energy remain nearly the same across all aluminium concentrations despite the fact that the activity coefficient of the solvent (Fe) was differently represented in both the approaches. Nevertheless, it is preferable to use the R-K polynomial due to its mathematical simplicity.
5. The chemistry and solution behaviour of the binaries (Fe-O and Fe-Al) and ternary (Fe-Al-O) systems, have been described. In general, the secondary steelmaking process of steels containing high aluminium has three stages: (i) before deoxidation, (ii) deoxidation with aluminium, and (iii) an alloying stage with high aluminium concentration. Consequently, these three stages can be linked to Fe-O, Fe-Al-O, and Fe-Al systems respectively.

CRediT authorship contribution statement

Mir Ishfaq: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Manish M. Pande:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [Manish M Pande reports financial support was provided by Science and Engineering Research Board (SERB), Department of Science and Technology, Government of India. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper].

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Data availability

Data will be made available on request.

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